

Personal Research Project Proposal

AI-Agent-Based Competitor Intelligence and Customer Sentiment Analysis for Laptops

Project Description

This project proposes the development of an AI-driven competitor intelligence system for the laptop market. The market for laptops involves strong competition between dozens of companies like Dell, HP, Lenovo, ASUS, Acer, Apple, MSI, Razer, and many others providing a wide variety of products targeting different use cases such as ultrabooks, gaming laptops, business workstations, 2-in-1 devices, and affordable models. For this reason, the market can be used to create a competitive intelligence system.

The proposed system will receive the chosen laptop as an input and find similar products by leveraging embedding-based similarity search, feature engineering, and retrieval-based ranking. Moreover, the system will use natural language processing (NLP) to analyze the feedback provided by customers of competing products. The analysis will involve sentiment analysis, theme extraction, and aspect-level review mining. In doing so, it will be possible to understand what drives customer satisfaction and dissatisfaction.

The AI agent will be designed with an intuitive user interface using Streamlit that will enable end-users to make queries on any product and receive answers regarding their competitors, prices, reviews, and market insights in general. For purposes of enhancing the performance and accuracy of the solution proposed here, we will be using a Retrieval-Augmented Generator framework in which generated texts should be grounded in terms of the retrieved metadata and reviews related to the product. This is crucial in mitigating issues such as hallucinations.

Data Source and Preparation

For this task, Amazon laptops dataset on Kaggle would be used, including structured information about the laptop such as brand, model name, processor, RAM, storage, screen size, operating system, price, and rating, along with reviews posted by customers. Amazon laptop products are some of the most popular products that receive thousands of reviews each day, making it possible to develop an effective competitive product matching module and review intelligence module.

The data preparation process will involve the following steps:

1. Deduplication and normalizing product listings to ensure there is only a single record of the same product with variants in terms of configuration, such as laptops sold with different amount of RAM, storage space, etc.
2. Filtering to retain only laptop products and exclude peripherals, accessories, laptop bags, and unrelated items that may appear in the dataset.

3. Parsing and structuring of free-text specifications (e.g., identifying CPU generation, graphics card type, and screen resolution from product descriptions)
4. Handling missing values in price, rating, and specification fields through imputation or exclusion based on missingness thresholds.
5. Preprocessing review text through lowercasing, removal of HTML artifacts, tokenization, and stopword filtering to prepare it for NLP modeling.
6. Constructing feature vectors at the product level that incorporate structured data from specification information as well as unstructured data from textual embeddings for similarity search.

In the event that the Amazon dataset is inadequate in terms of the level of detail provided by the specification information, additional information could be sourced from external databases, including Noteb.com and the Laptop Prices database on Kaggle.

3. Analytical Challenges

This project addresses several interconnected technical and analytical challenges:

1. **Competitor identification.** Laptops differ across multiple dimensions such as brand, processor line and series, GPU line and generation, RAM, storage type and capacity, screen size and resolution, weight, and intended purpose (game, ultrabook, business, budget). In contrast to straightforward cases of competitor matching, laptops require the careful balance of several attributes due to the mismatched price-based pairing of a low-budget Chromebook with an already discounted business laptop.
2. **Multimodal data integration.** It is necessary to merge structured information about specifications (processors, RAM capacity, price, ratings) with unstructured reviews into a single representation of products.
3. **Specification parsing and standardization.** Specification parsing and standardization. Specifications in laptops are often included in text-form product descriptions but not organized into clearly defined fields. Parsing and normalizing "Intel Core i7-13700H" vs. "i7 13th Gen" or "NVIDIA RTX 4060" vs. "GeForce 4060 GPU" needs special attention
4. **Review noise and subjectivity.** Customer comments are inconsistent and subjective. Laptops usually contain numerous aspects such as performance, screen quality, keyboard comfort, battery life, and even client service within one review which complicates its analysis compared to other simpler product classes.
5. **Hallucination risk in generative outputs.** When using a generative module for tasks such as summarization or answering questions, it is essential to reduce the risk of hallucinations by implementing techniques like retrieval from real data, generating answers based on evidence, and incorporating sources in the output.
6. **Decision-support translation.** Transforming analytical outputs into a user-friendly format that enables competitor benchmarking and insight formation becomes critical because the number of dimensions in laptop specifications is extremely high.

4. Scope and Prioritization

Given the extensive nature of the system, the project has been structured with deliverables and stretch goals. Deliverables are the basic minimum system and constitute the core aspect of the project schedule. The stretch goals add a layer of analysis and will be taken up if time allows.

Core Deliverables (Primary Focus)

Component	Description
Data pipeline	Cleaned, structured dataset of laptops and customer reviews from the Amazon Laptops dataset, with parsed and normalized specifications
Competitor matching	Embedding-based similarity search and retrieval-ranked product matching across specs, price, and use-case segment
Pricing benchmarking	Price-positioning analysis and comparison module for competing laptops within and across segments
Sentiment analysis	Review-level sentiment classification and aspect-level theme extraction (e.g., performance, display, battery, build quality)
Streamlit application	Interactive web interface for product query, comparison, and insight display
RAG-based AI agent	Natural language Q&A interface grounded in retrieved product specs and review evidence
Final report	Methodology, validation, findings, limitations, and business implications

Stretch Goals (If Time Permits)

Component	Description
Hallucination mitigation	Evidence-linking framework with source attribution and confidence scoring
Segment-level intelligence	Cross-segment analysis comparing competitive dynamics across gaming, ultrabook, business, and budget categories

The hallucination Mitigation and Segment-Level Intelligence fall under the stretch goals of the project as they require the completion of all other core deliverables and bring an extra level of difficulty when creating prompts and retrieving information.

5. Evaluation Plan

Each module of the system will be analyzed based on specific criteria for determining its reliability and validity.

Competitor Matching

Laptops that represent a variety of categories (gaming, ultrabook, business, economy) will be selected manually and checked to determine if the top-k returned competitors make a reasonable match. Precision at k ($P@k$) measure will be used to assess performance relative to

the ground truth test set of 40-60 laptop pairs manually labeled by a human. Ranking will be evaluated by means of nDCG.

Sentiment Analysis

The sentiment classification accuracy will be validated by means of comparing the sentiment prediction to the review rating (e.g., ratings from 1 to 2 stars = negative sentiments, ratings from 4 to 5 stars = positive sentiments). The accuracy, precision, recall, and F1 score will be used for assessing the sentiment classifier performance. Regarding aspect-based thematic information extraction, a qualitative assessment of the extracted aspects (e.g., battery life, keyboard quality, screen brightness) in 50 review excerpts will ensure their correctness.

Pricing and Benchmarking

The price function will be analyzed using a comparison of system-generated price data against manually generated price data. The statistics (brand average price and price dispersion within segments) will be checked for consistency against the original data set.

RAG Agent

If Evaluation criteria of the RAG model would include answer faithfulness (the degree to which the response is supported by the retrieved content), as well as answer relevance (the degree to which the response pertains to the question posed by the user). A total of 20 test questions will be created, including those pertaining to specifications comparison, product recommendations based on reviews, and industry-level questions. The scores will be calculated on a scale of 1 to 5 for each criterion separately. The hallucination rate will be calculated based on the percentage of unsupported responses.

6. Expected Outcomes

The anticipated output would be an AI-driven decision-making tool that allows the user to assess the competitiveness of a chosen laptop. In relation to a particular laptop, the system would provide a list of the closest competitors, perform comparisons according to price and performance on the market, determine customer sentiments, and extract the recurring positive and negative issues related to various attributes such as performance, screen, battery, build quality, etc.

On a larger scale, the project is aimed at delivering information about the price positioning across different categories, performance patterns within the brands, recurring issues of customers such as overheating, flexing keyboards, bad trackpads, among others, as well as factors that influence consumer perception within individual specifications.

7. Deliverables

The project will produce the following deliverables:

1. A cleaned and structured dataset of laptop products and customer reviews, sourced from the Amazon Laptops dataset on Kaggle, with parsed and normalized hardware specifications.

2. A product representation and competitor matching pipeline using embeddings and similarity-based retrieval across specifications, price, and use-case segment.
3. A pricing and benchmarking module for comparing competing laptops on price positioning and marketplace performance.
4. A sentiment analysis and review intelligence pipeline for classifying customer feedback and extracting aspect-level themes.
5. A Streamlit-based interactive web application for product comparison and market analysis.
6. An evaluation report documenting the performance of each component against the criteria defined in Section 5.
7. A final project report describing the methodology, model design, validation approach, findings, limitations, and business implications.
8. A prototype of an AI agent using the RAG model, evidence-linked outputs, a hallucination mitigation system, and segment-level competitive intelligence.

8. Analytical Skills and Learning Objectives

This project reinforces learning objectives across four analytical domains:

- **Machine Learning:** product similarity modeling, embedding-based retrieval, ranking design, and specification-aware feature engineering.
- **Text Analytics:** NLP-based sentiment analysis, topic extraction, aspect-level mining across multi-dimensional review content, and review summarization.
- **Marketing Analytics:** competitor benchmarking, price-positioning analysis across market segments, and brand-level intelligence generation.
- **Predictive and Diagnostic Analytics:** identifying specification and experience factors associated with stronger or weaker customer response across products, brands, and segments.

In addition, the project will develop technical skills in interactive analytics deployment via Streamlit, data-driven communication of insights, RAG system design, AI-agent workflow engineering, and cross-segment market analysis.